MILITARY SPECIFICATION

CLOTH, NAINSOOK, COTTON

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers the requirements for a cotton nainsook cloth.

2. APPLICABLE DOCUMENTS

2.1 Government documents

*2.1.1 Specifications, standards and handbooks Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified.

SPECIFICATIONS

FEDERAL

PPP-P-1134	- Packaging and Packing of Cotton and Cotton Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

STANDARDS

FEDERAL

FED-STD-4	- Glossary of Fabric Imperfections
FED-STD-191	- Textile Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

*2.1.2 Other Government documents, drawings, and publications The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

RULES AND REGULATIONS UNDER THE TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Application for copies should be addressed to the Federal Trade Commission, Washington, DC 20580).

*2.2 Other publications The following document(s) form a part of this specification to the extent specified herein. The issues of documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and supplement thereto, if applicable.

American Society of Testing and Materials (ASTM) D 1424-63 Test for Tear Resistance of Woven Fabrics by Falling Pendulum (Elmendorf) Apparatus

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

3. REQUIREMENTS

3.1 Standard sample The finished cloth shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article When specified (see 6.2), the contractor shall furnish a sample of cloth for first article inspection and approval.

3.3 Material (see 6.4).

3.3.1 Yarn The yarn shall be made of cotton which has been cleaned, carded, drawn and spun into singles yarn for both the warp and the filling when tested as specified in 4.5.

3.4 Physical requirements The finished cloth shall conform to the requirements in Table I when tested as specified in 4.5.

Table I - Physical requirements

Characteristics	Requirement	
	Minimum	Maximum
Weight, ounces per square yard (g/m^2)	2.8 (94.9)	3.5 (118.7)
Yarns per inch (2.5cm)		
Warp	80	100
Filling	74	100
Breaking strength, pounds (Newtons)		
Warp	50 (222.51)	-
Filling	30 (133.5)	-

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Table I - Physical requirements (cont'd)

Characteristics	Requirements	
	Minimum	Maximum
Tearing strength, grams (Newtons)		
Dry		
Warp	880 (8.6)	-
Filling	480 (4.7)	-
Wet-water		
Warp	1040 (10.2)	-
Filling	720 (7.1)	-
Wet-acetone		
Warp	560 (5.5)	-
Filling	320 (3.1)	-

3.4.1 Weave The weave shall be plain weave.

*3.4.2 Width The width shall be as specified (see 6.2) and shall be the minimum acceptable width, inclusive of selvage. When the cloth is manufactured with shutteless loom selvages, the width of the cloth should be as specified (see 6.2) and shall be the minimum width of the woven cloth, excluding the selvages.

*3.4.3 Length and put-up Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards (36.4m). Each length shall be put up on full width rolls as specified in PPP-P-1134, with the fabric face on the inside.

3.5 Finish The cloth shall be bleached and shall have a clear finish and hand equal to that of the standard sample.

3.5.1 pH The pH value of the water extract of the finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.5.

3.5.2 Nonfibrous materials The finished cloth shall not exceed 2.0 percent starch and protein content including chloroform-soluble and water-soluble material when tested as specified in 4.5.

3.6 Absorbency The cloth shall be considered absorbent when the water and oil are absorbed into the fabric within 30 seconds, when tested as specified in 4.5.

*3.7 Fiber identification Each roll of cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.8 Marking The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "face" at each end of the piece.

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3.9 Workmanship The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards, when calculated as specified in Section 4, shall not excced the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

*4.2 Classification of inspection The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3)
2. Quality conformance inspection (see 4.4)

4.3 First article inspection When required, the first article of finished cloth shall be visually inspected for color, finish and appearance and shall be tested for chemical and physical properties in accordance with the applicable methods specified in 4.5.

*4.4 Quality conformance inspection Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 Examination of the end item Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.3.

4.4.2.1 Yard-by-yard examination Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards (91.4m), the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards (91.4m), only 100 yards (91.4m) shall be examined. All defects as defined in Section I of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) (0.9m), shall be scored and assigned demerit points as listed in 4.4.2.1.1, except that only those slubs and knots which exceed the limit shown in Figure 1 of FED-STD-4 shall be scored. No linear yard (increment of 1 yard (0.9m) on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample

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size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards ($83.6m^2$) of the total yardage examined excees 35.0 points. The lot shall be unacceptable if the points per 100 square yards ($83.6m^2$) of two or more individual rolls exceed 48.0 points. If one roll exeeds the point level, a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceed 48.0 points per 100 square yards ($83.6m^2$). Point computation for quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample X 3600}}{\text{Contracted width of cloth (inches) X total yards inspected}} = \text{Points per 100 square yards}$$

4.4.2.1.1 Demerit points Demerit points shall be assigned as follows:

For defects up to and including 3 inches (7.6cm) in any dimension - one point
For defects exceeding 3 inches (7.6cm), but not exeeding 6 inches (15.2cm) in any dimension - two points
For defects exceeding 6 inches (15.2cm) but not exceeding 9 inches (22.9cm) in any dimension - three points
For defects exceeding 9 inches (22.9cm) in any dimension - four points

The following defects, when present, shall be scored 4 points for each yard (0.9m) in which they occur:

Baggy, ridgy or wavy cloth
Width less than specified
Excessive neppiness

4.4.2.2 Examination for length

4.4.2.2.1 Individual rolls During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified, or more than two yards (1.8m) less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.4.2.2.2 Total yardage in sample The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for the examination of individual rolls.

*4.4.3 Examination for face identification and nonconformance with the Textile Fiber Products Identification Act During the yard-by-yard examination, each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample have face identification missing from either or both ends or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

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4 4.4 Examination of packaging requirements An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing and marking comply with the Section 5 requirements of this specification.

4.5 Testing of the end item The methods of testing specified in FED-STD-191, wherever applicable and as listed in Table II, shall be followed. The physical and chemical values specified in Section 3, except where otherwise specified, apply to the average of the determinations made on the sample unit for test purposes as specified in applicable test methods. All test reports shall contain the individual values utilized in expressing the final results. The sample unit for test purposes shall be 2 1/2 continuous yards (2.3cm) full width of the finished cloth. The lot size shall be expressed in units of one yard. The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Table II - Test methods

Characteristics	Requirement paragraph	Test Method
Yarn identification	3.3.1	1200 3/
Weight	3.4	5041
Yarns per inch		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength		
Warp	3.4	5100
Filling	3.4	5100
Tearing strength		
Dry		
Warp	3.4	ASTM D 1424-63
Filling	3.4	ASTM D 1424-63
Wet (water)		
Warp	3.4	ASTM D 1424 -63 1/
Filling	3.4	ASTM D 1424 -63 1/
wet (acetone)		
Warp	3.4	ASTM D 1424 - 63 1/
Filling	3.4	ASTM D 1424 - 63 1/
Finish	3.5	3/
Weave	3.4.1	Visual 2/
Nonfibrous material	3.5.2	2611
Absorbency	3.6	4.5.1
pH	3.5.1	2811

1/ The test samples shall be cut the appropriate size for testing and immersed in distilled water or acetone, as applicable, for one minute. Remove and place between two sheets of blotting paper conforming to the type specified in FED-STD-191, TM 5500, and pass through a household type wringer exerting a total load on the specimen of approximately 60 pounds. The specimens shall be immediately tested for tearing strength.

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2/ One determination shall be made from each sample unit and the results reported as "pass" or "fail".
3/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirements.

4.5.1 Test for absorbency The specimen shall be spread out on a non-absorbent surface. Drops 0.04 milliliter in volume of water and light machine oil shall be applied separately to the sample cloth through a capillary by allowing the drops to fall continuously and freely onto the sample from a height of 1/2 to 2 inches. After a time interval of 30 seconds, the sample shall be examined and the absorption and non-absorption of the water and oil reported. The cloth is considered non-absorbent if the water or oil is not absorbed but remains wholly or partly above the surface of the fabric or the supporting surface.

5. PACKAGING

5.1 Preservation-packaging Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C The cloth, put up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

6. NOTES

6.1 Intended use The cloth covered by this specification is intended for use as a filter cloth and in fabricating Grade A cleaning cloths.

6.2 Ordering data Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Width required (see 3.4.2).
c. When length is other than specified (see 3.4.3).
d. Levels of packaging and packing required (see 5.1 and 5.2).
*e. When a first article inspection is required, the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, quantity, and testing and approval of the first article.

6.3 Standard sample For access to the standard sample, address the procuring activity issuing the invitation for bids.

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*6.4 Recycled material It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

*6.5 Changes from previous issue The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evalute the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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